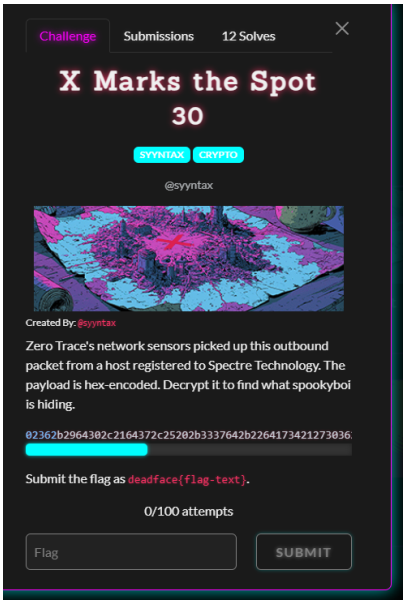
Solving “X Marks the Spot” (30 pts - Crypto)

A compact hex-decoding and single-byte XOR walkthrough.



Challenge page used during the solve.

1. The Challenge

The challenge provided a long hexadecimal string and said the payload was hex-encoded. After decoding, the bytes were still unreadable, so a second transformation was required.

2. Hex Decode

```
02362b2964302c2164372c25202b3337642b226417342127303621641021272c2a2b282b233d7e64202
12520222527213f3c2b361b751b263d30771b70301b701b3075297739
```

3. Test a Single-Byte XOR

XORing each decoded byte with 0x44 produces clean English immediately, which confirms the key.

```
From the shadows of Spectre Technology: deadface{xor_1_byt3_4t_4_t1m3}
```

4. The Flag

```
deadface{xor_1_byt3_4t_4_t1m3}
```

5. General Method

• Decode the outer representation first (hex -> bytes).

• When the result is high-entropy but short, test single-byte XOR keys.

• Score candidate plaintexts for printable English and known flag prefixes.

Summary

Field	Result
Category	Cryptography
Layers	Hex encoding + XOR
XOR key	0x44
Flag	deadface{xor_1_byt3_4t_4_t1m3}